OPTIMAL CONTRACT AND GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: EXISTENCE AND UNIQUENESS, A UNIFORM HOMOTOPY CONDITION, THE GENERAL-F EMISSION KEYSTONE, CONNECTEDNESS, AND SHARPNESS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

ABSTRACT. This is the Role 1 Week 2 deliverable. It hardens the Week 1 carbon theory (`role1_carbon_theory.tex`) into five results—four proved, one resolved numerically—taking the same model and notation and staying in Role 1's large-FCL mandate (the one small-FCL touch is flagged as a conjecture pending Role 4). *One*, the regulated program has a maximizer, it is unique, and the within-period dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$ is the unique optimal rule; uniqueness comes not from global concavity—which we neither assume nor have, since it lapses on the power-law knife-edge where the reduced renewable problem goes flat—but from a *recursive-triangular* structure—the distribution-free cutoffs pin K_F and then K_M as unique functions of K_R, after which a single scalar diagonal-dominance (reduced second-order) condition pins K_R. *Two*, support-wide monotonicity of the shift ratio $g = F_{K_R}/f$ makes the realized two-point endpoint inequality hold *uniformly* along the whole C_M-homotopy, even as the dispatch window moves—the exact content the Week 1 "integrate-downward" step needs. *Three* (the keystone), substituting the equilibrium map $K^*(C_R)$ into the exact inequality (C) yields the endogenous carbon condition, and we give a *primitive* sufficient condition—the endpoint shift-ratio condition plus a two-zone density (tail/skew) bound—that would make carbon fall at a *finite interior* optimum. We then settle whether it ever fires, with two *proved* results and a numerical benchmark. First, a fossil floor: $\mathbb{E}[\text{fossil}^*] \geq K_F^*(R - c - C_M)/(R - c)$ (and $\geq K_M^* - K_R^*$), since an FCL exceeding the maximal renewable surplus $\leq K_R^*$ cannot be fully surplus-powered. Second—the key structural point—we prove the carbon sign is *irreducibly distributional*: holding the equilibrium capacities and cutoffs fixed, the admissible conditional laws make Δ range over an interval that straddles zero, so capacities and cutoffs together cannot determine $\operatorname{sign}\Delta$ and Week-1's conjecture admits no proof from the distribution-free structure alone. The residual question is then the sign of a cutoff-computable benchmark Δ_0 plus a density tilt; numerically, at every large-FCL interior optimum scanned $\Delta_0 < 0$ ($\Delta_0^{\max} \approx -1.09$) and the tilt reinforces it, $\Delta < 0$ ($\Delta^{\max} \approx -1.40$). Carbon strictly falls only in the cheap-renewable corner $K_R^* \to \infty$. So the honest reading is *negative*: interior carbon reduction is a corner phenomenon (the small-FCL regime is Role 4's, not claimed here). *Four*, the carbon-reduction set $\{C_R : \Delta(C_R) \geq 0\}$ is bounded above and non-empty—its non-emptiness coming from that power-law corner, not from an interior optimum—and is a single interval under an explicit monotone-renewable-shadow-value condition that, we note, already encodes the single-crossing property it certifies. *Five*, the sharp necessary-and-sufficient condition for $K_R^* > K_R^\circ$ is the *path integral* of the realized endpoint gap; pointwise-monotone g is sufficient but not necessary; and the two-sided bounds on the carbon saving are tight, attained by explicit degenerate conditional laws. Every identity is checked against quadrature of the profit integral in `verify_week2.py`.

1. SETUP

We keep the model and notation of `role1_carbon_theory.tex`. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \geq 0$, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Date: Working results, Week 2.

Assumption 1 (Regularity, as in Week 1)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \beta + p > R.$

Under the dispatch rule $d(x) = \min\{(K_F - x)_+, K_M\}$ the unified profit is

$$\Pi = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \quad (1)$$

with gradient (Week 1, Lemma 1)

$$g_M := \Pi_{K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \quad (2)$$

$$g_F := \Pi_{K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \quad (3)$$

$$g_R := \Pi_{K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \quad (4)$$

We work in the *large-FCL interior* regime $K_M \geq K_R$ (so $F(-K_M) = f(-K_M) = F_{K_R}(-K_M) = 0$), the regime in which the Week-1 carbon theory lives; throughout, "interior" means $0 < K_R < \infty$ and the cutoffs below sit strictly inside $(0, 1)$. Write $a := K_F - K_M$ for the lower dispatch endpoint and recall the distribution-free cutoff *values* (Week 1, Lemma 2)

$$\varphi_1 := F(a; K_R) = \frac{C_M}{R-c}, \qquad \varphi_2 := F(K_F; K_R) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}, \qquad \varphi^\circ := F(K_F^\circ; K_R^\circ) = \frac{\beta+p-c-C_F}{\beta+p-c} \quad (5)$$

Under Assumption 1, $0 < \varphi_1 < 1$ and $0 < \varphi_2 < 1$; the interior FCL (adoption) condition $K_M > 0$ is $\varphi_2 > \varphi_1$, equivalently $C_M < \bar{C}_M := (R-c)\varphi^\circ$.

2. Existence, uniqueness, and contract optimality

The optimum is not delivered by global concavity, which we neither assume nor establish. The Week-1 cross-partial $\Pi_{K_R K_F} = (R - \beta + p)F_{K_R}(K_F) - (R-c)F_{K_R}(a) \leq 0$ shows Π is *submodular* in (K_R, K_F)—a feature, not yet an obstruction (a submodular function can still be concave). The real point is that concavity genuinely *fails* on the power-law knife-edge: there the reduced renewable problem Ψ of (7) is flat ($\Psi' \equiv 0$, Condition 1 at equality), so Π is not strictly concave along the renewable direction and the naive route of signing the full 3×3 Hessian by inspection is unavailable. Uniqueness instead follows from a recursive-triangular structure together with a single scalar condition.

2.1. Dispatch optimality.

Lemma 1 (Unique optimal dispatch via strict $R > c$)**.** *Fix capacities (K_R, K_F, K_M) and a residual-demand realization x. Consumers are served first (they outrank FCL for scarce fossil because $\beta + p - c > R - c$): fossil generation $\min\{(x)_+, K_F\}$ meets consumer residual demand, leaving fossil headroom $(K_F - (x)_+)_+$, while renewable surplus $(-x)_+$ is free power. The FCL dispatch $d \in [0, K_M]$ is therefore feasible up to $d \leq (-x)_+ + (K_F - (x)_+)_+ = (K_F - x)_+$, and the fossil it burns is $(d - (-x)_+)_+$ (surplus first, then headroom). Its within-period payoff is*

$$\pi_x(d) = R\,d - c\,(d - (-x)_+)_+, \qquad 0 \leq d \leq \min\{K_M, (K_F - x)_+\}.$$

Under Assumption 1 ($R > c$) the unique maximizer is $d^*(x) = \min\{(K_F - x)_+, K_M\}$.

Proof. π_x is continuous, piecewise linear, and strictly increasing in d: on the surplus segment $d \leq (-x)_+$ the slope is $R > 0$ (FCL runs on free renewable power), and on the fossil segment $d > (-x)_+$ the slope is $R - c > 0$ (Assumption 1). Both slopes are strictly positive, so π_x is maximized at the right endpoint $d^* = \min\{(K_F - x)_+, K_M\}$, and uniquely so. At $R = c$ the fossil-segment slope vanishes and any $d \in [(-x)_+, \min\{K_M, (K_F - x)_+\}]$ is optimal; hence strict $R > c$ is exactly what pins d^*. Crucially the feasibility $d \leq (K_F - x)_+$ lets surplus states $x < 0$

power the FCL beyond fossil headroom K_F—the channel the floor of Lemma 5 quantifies. This d^*, with consumer fossil $\min\{(x)_+, K_F\}$, reproduces the operating-cost terms of (1).[1] □

2.2. **Existence.**

Proposition 1 (A maximizer exists)**.** Π *is continuous on* $\mathbb{R}^3_+$*, and there is a box* $[0, \bar{K}]^3$ *that no maximizing sequence leaves. Hence the set of maximizers is nonempty and compact.*

Proof. Continuity is immediate: each integrand in (1) is bounded and $F(\cdot;\cdot)$ is jointly continuous. We show Π is *coercive*— $\Pi(K) \to -\infty$ as $\|K\| \to \infty$—which with continuity gives a maximizer on a compact sublevel set by Weierstrass. Since $\tilde{x}$ has bounded support $[-K_R, \bar{D}]$ ($F \equiv 0$ below $-K_R$, $F \equiv 1$ above $\bar{D}$), each operating integral grows at most linearly in $\|K\|$, and the blackout term $-(\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx \le 0$ only helps; so it suffices to balance the FCL revenue and surplus-running cost against capital along every ray $K = t\,u + O(1)$, $u \ge 0$, $u \ne 0$, $t \to \infty$.

Axis rays. For $u = e_M$ the band $[K_F - K_M, K_F]$ slides below the support, so $\int_{K_F-K_M}^{K_F} F\,dx$ and $\int_{-K_M}^{K_F-K_M}(1-F)\,dx$ saturate to constants and $\Pi = -C_M K_M + O(1) \to -\infty$; $u = e_F$ gives $-C_F K_F + O(1)$; and $u = e_R$ gives $-C_R K_R + O(1)$ (with $0 \le \int F_{K_R}\,dx = \mathbb{E}[\tilde{S}] < \infty$ over the fixed windows). Each axis ray is coercive from its own capital cost—so the costs $C_R, C_F, C_M > 0$ already kill the single-capacity rays.

Mixed K_F, K_M rays. Put $K_F = \gamma K_M + O(1)$, $\gamma > 0$, K_R bounded. Counting only the in-support part of each band (where F rises from 0 to 1), $\int_{K_F-K_M}^{K_F} F\,dx = \min\{1,\gamma\}\,K_M + O(1)$ and $\int_{-K_M}^{K_F-K_M}(1-F)\,dx = \min\{1,\gamma\}\,K_M + O(1)$, so the K_M-coefficient of Π is

$$N(\gamma) = (R-c)\min\{1,\gamma\} - \gamma C_F - C_M.$$

For $\gamma \ge 1$, $N = (R-c) - \gamma C_F - C_M \le (R-c) - C_F - C_M < 0$; for $\gamma < 1$, $N = \gamma(R-c-C_F) - C_M < \gamma C_M - C_M \le 0$ (using $R - c - C_F < C_M$). Both use exactly the binding hypothesis $R < c + C_F + C_M$ (Assumption 1): without it the FCL revenue could outrun the fossil it burns. Allowing $K_R \to \infty$ too adds $-C_R K_R \le 0$ and cannot raise N. Hence $\Pi \to -\infty$ along every ray, so Π is coercive and a maximizer exists.[2] □

2.3. **Uniqueness by recursive triangularity.** Fix K_R. By (5) the equations $g_M = 0$, $g_F = 0$ are equivalent to $F(a; K_R) = \varphi_1$ and $F(K_F; K_R) = \varphi_2$, two *distribution-free* targets. Since $x \mapsto F(x; K_R)$ is a strictly increasing bijection from $(-K_R, \bar{D})$ onto $(0,1)$, each target has a unique preimage:

$$K_F(K_R) = F^{-1}(\varphi_2; K_R), \qquad a(K_R) = F^{-1}(\varphi_1; K_R), \qquad K_M(K_R) = K_F(K_R) - a(K_R) > 0, \tag{6}$$

all C^1 in K_R (implicit function theorem, $\partial F/\partial x = f > 0$). This is the triangle: levels one and two (K_F, K_M) are pinned by K_R with no distributional input. Substituting (6) into g_R defines the reduced renewable condition

$$\Psi(K_R) := g_R\big(K_R, K_F(K_R), K_M(K_R)\big), \qquad \Psi(K_R^*) = 0. \tag{7}$$

Lemma 2 (The (K_F, K_M) block is negative definite given interiority)**.** *At any large-FCL interior point (so $f(a), f(K_F^*) > 0$, the cutoffs strictly inside the support) the 2×2 Hessian block in (K_F, K_M), $\mathcal{M} = \begin{pmatrix} \Pi_{K_F K_F} & \Pi_{K_F K_M} \\ \Pi_{K_F K_M} & \Pi_{K_M K_M} \end{pmatrix}$, satisfies $\Pi_{K_F K_F} < 0$, $\Pi_{K_M K_M} < 0$, and*

$$\det \mathcal{M} = (R-c) f(a)\,(\beta + p - R) f(K_F) \;>\; 0,$$

so $\mathcal{M} \prec 0$ under Assumption 1 given interiority (no distributional *condition beyond $f > 0$ at the cutoffs).*

[1] Verified in `verify_week2.py` T1a: a grid search of $\pi_x(d)$ over the feasible interval returns d^* at every sampled x, including surplus states where $d^* > K_F$.

[2] T1b: the own partials are strictly negative at $K_i = 60$ (others moderate), and the mixed-ray coefficient $N(\gamma) < 0$ is checked on a γ-grid; the binding role of $R < c + C_F + C_M$ is confirmed by flipping it.

Proof. From (2)–(3) in the large-FCL regime, $\Pi_{K_M K_M} = -(R-c)f(a) < 0$, $\Pi_{K_F K_M} = (R-c)f(a)$, and $\Pi_{K_F K_F} = (R-\beta+p)f(K_F) - (R-c)f(a) < 0$ (both terms negative since $R < \beta + p$, $R > c$). Then $\det \mathcal{M} = \Pi_{K_F K_F}\Pi_{K_M K_M} - \Pi^2_{K_F K_M} = [-(R-c)f(a)]\,(R-\beta+p)f(K_F) + (R-c)^2 f(a)^2 - (R-c)^2 f(a)^2 = (R-c)f(a)(\beta+p-R)f(K_F) > 0$. Negative diagonal and positive determinant give $\mathcal{M} \prec 0$.[3] □

Condition 1 (Reduced second-order / diagonal dominance)**.** Along the interior locus (6), the Schur complement of the renewable coordinate is negative,

$$\Psi'(K_R) = \Pi_{K_R K_R} - \begin{pmatrix} \Pi_{K_R K_F} & \Pi_{K_R K_M} \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \Pi_{K_R K_F} \\ \Pi_{K_R K_M} \end{pmatrix} < 0. \tag{8}$$

Condition 1 is a *single scalar* inequality—the second-order condition of the one-dimensional reduced problem $\Psi(K_R) = 0$, i.e. negative diagonal dominance of the Hessian in the renewable row after the (K_F, K_M)-block has been eliminated. It is strictly weaker than global concavity (which submodularity rules out) and it is exactly what lapses, becoming an equality, at the power-law knife-edge where $\Psi \equiv$ const. *Primitive content.* $\Psi(K_R) + C_R$ is the marginal value of renewables along the re-optimization locus; $\Psi' < 0$ says this marginal value is strictly decreasing in K_R—the renewable problem is single-peaked along the locus. The power-law family is the exact boundary ($\Psi' \equiv 0$, constant marginal value: the Week-1 knife-edge result for the power-law family), so any family with strictly diminishing renewable returns along the locus—which we verify for the Beta scale mixtures by the eigenvalue scan of §7—satisfies it strictly. (We give a checkable scalar, not a closed-form primitive on F; deriving $\Psi' < 0$ from a density condition on F in general is open.)

Theorem 1 (Existence, uniqueness, optimality)**.** *Under Assumption 1 and Condition 1, the profit* Π *has* at most one *interior critical point; it exists whenever the program's optimum is interior (the cutoffs* (5) *lie strictly in* $(0,1)$ *with* $K_M > 0$*—the Week-1 large-FCL adoption regime,* $C_M < \bar{C}_M$ *and* C_R *not so large that* $K_R^* = 0$*), in which case it is the* unique *interior maximizer* (K_R^*, K_F^*, K_M^*)*, the Hessian is negative definite there, and* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal dispatch.*

Proof. Any interior critical point solves $\nabla\Pi = 0$, hence (cutoff Lemma) lies on the locus (6) and satisfies $\Psi(K_R) = 0$. By (7) and the chain rule, $\Psi'(K_R)$ equals the Schur complement (8) (the total derivative of g_R along the locus, after eliminating (K_F, K_M) through $g_M = g_F = 0$). Lemma 2 gives $\mathcal{M} \prec 0$, so $\mathcal{M}^{-1} \prec 0$ and the quadratic form in (8) is ≥ 0; Condition 1 makes $\Psi'(K_R) < 0$, so Ψ is strictly decreasing and has *at most one* zero—hence at most one interior critical point. When the optimum is interior (the stated regime), a maximizer exists by coercivity (Proposition 1) and lies in the interior, so it is a critical point and therefore *the* unique one K_R^*; (6) returns unique K_F^*, K_M^*. By the Schur determinant identity $\det \mathcal{H} = \det \mathcal{M} \cdot \Psi'(K_R)$ with $\det \mathcal{M} > 0$ and $\Psi' < 0$, the leading-principal-minor signs alternate, so $\mathcal{H} \prec 0$ at the optimum; the dispatch claim is Lemma 1.[4] □

Remark 1. The triangle is the whole point. Concavity would ask the full 3×3 Hessian to be negative definite as a joint hypothesis; the recursive structure instead *derives* negative definiteness from two cheap facts—an interiority-only 2×2 block (Lemma 2) and one scalar slope (Condition 1)—which is why the submodularity in (K_R, K_F) is no obstruction.

Corollary 1 (The equilibrium map is C^1)**.** *On the open large-FCL interior region where Condition 1 holds, the optimum* (K_R^*, K_F^*, K_M^*) *is a* C^1 *function of the parameters* (C_R, C_M)*; likewise the no-FCL optimum* (K_R°, K_F°)*. Consequently* $\mathbb{E}[\textit{fossil}^*]$*,* $\mathbb{E}[\textit{fossil}^\circ]$*, and* $\Delta = \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ *are* C^1 *in* C_R *there.*

[3] T1c: $\det \mathcal{M}$ matches the closed form and $\operatorname{eig} \mathcal{M} < 0$ at the solved optimum, on Beta$(2,2)$ and Beta$(2,3)$ scale mixtures.

[4] T1c: $\Psi(K_R)$ is strictly decreasing on a scan of the locus with its unique zero at the solved K_R^* ($|\Psi(K_R^*)| < 10^{-6}$), and $\operatorname{eig} \mathcal{H} < 0$.

Proof. By Theorem 1, $\mathcal{H} \prec 0$ at the optimum, so the Jacobian of the C^1 map $K \mapsto \nabla\Pi(K; C_R, C_M)$ in K is nonsingular; the implicit function theorem gives $(C_R, C_M) \mapsto K_R^*, K_F^*, K_M^*$ in C^1. The no-FCL system $g_F = g_R = 0$ ($K_M \equiv 0$) has Jacobian $\mathcal{M}$'s no-FCL analogue, nonsingular by the same block argument, giving $K_R^\circ, K_F^\circ \in C^1$. Each fossil expectation is a smooth functional of the capacities, so $\Delta \in C^1$ by the chain rule.[5] □

3. The shift-ratio condition, uniform along the homotopy

Week-1 Theorem 1 signs $K_R^* > K_R^\circ$ by integrating $dK_R^*/dC_M \le 0$ downward from $\bar{C}_M$. We first record the comparative static it integrates, derived here so the homotopy argument is self-contained.

Lemma 3 (Capacity-channel cofactor identity)**.** *Write $g(x; K_R) := F_{K_R}(x; K_R)/f(x; K_R)$ and $a = K_F^* - K_M^*$. At a large-FCL interior optimum ($\mathcal{H} \prec 0$),*

$$\frac{dK_R^*}{dC_M} = \frac{\mathrm{cof}_{K_M,K_R}(\mathcal{H})}{\det \mathcal{H}}, \qquad \mathrm{cof}_{K_M,K_R}(\mathcal{H}) = (R-c)(\beta+p-R)\, f(a)\, f(K_F^*)\,[g(a) - g(K_F^*)]. \quad (9)$$

Since $\det \mathcal{H} < 0$ *and* $(R-c)(\beta+p-R)f(a)f(K_F^*) > 0$, $dK_R^*/dC_M \le 0$ iff $g(a) \ge g(K_F^*)$.

Proof. Only g_M depends on C_M, with $\partial g_M/\partial C_M = -1$, so differentiating $\nabla\Pi = 0$ gives $\mathcal{H}\,(dK/dC_M) = (0,0,1)^\top$ and, by Cramer's rule, $dK_R^*/dC_M = [\mathcal{H}^{-1}]_{K_R,K_M} = \mathrm{cof}_{K_M,K_R}(\mathcal{H})/\det\mathcal{H}$. The cofactor is the minor on rows (K_R, K_F), columns (K_F, K_M): $\Pi_{K_RK_F}\Pi_{K_FK_M} - \Pi_{K_RK_M}\Pi_{K_FK_F}$. Substituting the mixed partials $\Pi_{K_RK_M} = (R-c)F_{K_R}(a)$, $\Pi_{K_FK_M} = (R-c)f(a)$, $\Pi_{K_RK_F} = (R-\beta+p)F_{K_R}(K_F^*) - (R-c)F_{K_R}(a)$, and $\Pi_{K_FK_F} = (R-\beta+p)f(K_F^*) - (R-c)f(a)$ (the F_{K_R}-versions of Lemma 2's entries), the $(R-c)F_{K_R}(a)f(a)$ cross-terms cancel and

$$\mathrm{cof}_{K_M,K_R}(\mathcal{H}) = (R-c)(R-\beta+p)\, f(a)f(K_F^*)[g(K_F^*)-g(a)] = (R-c)(\beta+p-R)\, f(a)f(K_F^*)[g(a)-g(K_F^*)],$$

the second equality flipping both sign factors ($R-\beta+p = -(\beta+p-R)$). The sign claim follows from $\det\mathcal{H} < 0$. □

So the downward integration needs the *realized two-point endpoint inequality* $g(a) \ge g(K_F^*)$ to hold at *every* C_M on the path—even though the window $(a, K_F^*) = (K_F^* - K_M^*, K_F^*)$ moves with C_M. The following lemma supplies exactly that from a primitive, C_M-free hypothesis.

Condition 2 (Support-wide monotone shift ratio)**.** For each K_R in the equilibrium range, $x \mapsto g(x; K_R)$ is non-increasing on the support $(-K_R, \bar{D})$.

Lemma 4 (Uniform endpoint inequality along the homotopy)**.** *Suppose Condition 2 holds. Then for every $C_M \in (0, \bar{C}_M)$, at the interior optimum the window endpoints satisfy $-K_R^* < a(C_M) < K_F^*(C_M) < \bar{D}$ and*

$$g(a(C_M); K_R^*(C_M)) \;\ge\; g(K_F^*(C_M); K_R^*(C_M)), \quad (10)$$

with strict inequality wherever $g(\cdot; K_R^*)$ is strictly decreasing. Hence $\mathrm{cof}_{K_MK_R}(\mathcal{H}) \ge 0$ and $dK_R^*/dC_M \le 0$ uniformly on the path, and the Week-1 downward integration is justified by Condition 2 alone.

Proof. At an interior optimum with $C_M < \bar{C}_M$ the adoption margin is strict, $K_M^* > 0$, so $a = K_F^* - K_M^* < K_F^*$. The cutoffs (5) give $F(a; K_R^*) = \varphi_1 \in (0,1)$ and $F(K_F^*; K_R^*) = \varphi_2 \in (0,1)$, so both endpoints lie strictly inside the support $(-K_R^*, \bar{D})$ (a point where $F \in (0,1)$ cannot be at an extended-CDF plateau). Condition 2 applied at $K_R = K_R^*$ to the two ordered interior points $a < K_F^*$ yields $g(a; K_R^*) \ge g(K_F^*; K_R^*)$, which is (10). This holds for every C_M because the hypothesis is on the whole support, independent of where the window sits; as C_M sweeps $(0, \bar{C}_M)$ the window sweeps

[5] Smoothness of $K_R^*(\cdot)$ is exhibited numerically by the T5a fundamental-theorem-of-calculus test (its finite-difference derivative integrates back to the level on the interior band). The only non-smooth point is the large-/small-FCL crossing, which is outside this regime (Role 4).

a sub-family of ordered interior pairs, all covered at once. The cofactor and derivative signs then follow from Lemma 3.[6] □

Remark 2. Condition 2 is the honest primitive behind Week-1's shift-ratio condition (CS), which was stated on the moving window "at each point of the equilibrium path." Lemma 4 shows the moving-window statement is *implied* by the static support-wide one, so the comparative static rests on a hypothesis one can check on F without first solving for the path. The power-law family has $g(x) = (\bar{D} - x)/(\bar{D} + K_R)$, affine decreasing for every α, hence satisfies Condition 2 trivially.

4. The emission condition for general F (keystone)

Week 1 closed the carbon sign only on the power-law/uniform knife-edge (a *corner* $K_R^* \in \{0, \infty\}$) and left a non-empty strict-fall window for general F *conjectural*. We close it at a *finite interior* optimum. Recall (Week 1, Prop. 4) the exact net fossil saving $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ with

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}}), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}, \tag{11}$$

where $a = K_F^* - K_M^*$, $q^\circ = F(0; K_R^\circ)$, and $\mu^*_{\text{low}} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)]$, $\mu^\circ_{\text{mid}} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$. Carbon falls iff $\Delta > 0$, i.e. inequality (C) of Week 1 holds.

The endogenous condition. Substituting the equilibrium map $K_R^*(C_R), K_F^*(C_R), K_M^*(C_R)$ and $K_R^\circ(C_R), K_F^\circ(C_R)$ into (11) turns Δ into a function of primitives and C_R alone:

$$\Delta(C_R) = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}}\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}})\Big], \tag{12}$$

all objects evaluated at the equilibrium. The difficulty is that the conditional means $\mu^*_{\text{low}}, \mu^\circ_{\text{mid}}$ and the capacities move with the shape of F.

Condition 3 (Two-zone density / tail-skew bound)**.** At the equilibrium, (a) $f(\cdot; K_R^*)$ is non-increasing on the FCL zone $(-K_R^*, a)$, and (b) $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL zone $(0, K_F^\circ)$.

Part (a) is a left-tail/skew bound on *surplus* states (deep-surplus mass not too thin); part (b) a skew bound on the no-FCL mid zone. Both are monotone-density conditions on the model primitive f (the zone endpoints are equilibrium-determined, but one obtains a fully primitive sufficient version by requiring the monotonicity on the maximal zones swept as C_M, C_R range over the adoption region). Like Condition 2, these are conditions on F, not on the solved optimum.

Theorem 2 (General-F emission keystone)**.** *Suppose the program has a finite interior optimum (Theorem 1), Condition 2 holds (so $K_R^* > K_R^\circ$ by Lemma 4), and Condition 3 holds. Then*

$$\Delta \;\geq\; \Delta_{\text{LB}} := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] + \frac{C_M}{R-c}\cdot\frac{a + K_R^*}{2} - K_F^*, \tag{13}$$

and therefore carbon strictly falls ($\Delta > 0$) whenever $\Delta_{\text{LB}} > 0$. The bound is expressed in the equilibrium capacities and the distribution-free reliability φ° only. Caveat: *the condition $\Delta_{\text{LB}} > 0$ is a property of a capacity tuple, not a guarantee that any optimum attains it; Proposition 3 reports that it is essentially never attained at a solved interior optimum, so the value of* (13) *is its* form *(it pinpoints, via Lemma 5, the floor that defeats it), not its activation.*

Proof. Under Condition 3(a), a non-increasing density on $(-K_R^*, a)$ has conditional mean at most the interval midpoint, $\mu^*_{\text{low}} \leq (a - K_R^*)/2$ (the standard fact: a density non-increasing on $[\ell, u]$ has mean $\leq (\ell + u)/2$, with equality for the uniform); since (11) carries the term $-\frac{C_M}{R-c}(a - \mu^*_{\text{low}})$ (decreasing in μ^*_{low}), bounding $a - \mu^*_{\text{low}}$ below by $a - \frac{a - K_R^*}{2} = \frac{a + K_R^*}{2}$ gives the upper bound $\mathbb{E}[\text{fossil}^*] \leq K_F^* - \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$. Under Condition 3(b), a non-decreasing density on $(0, K_F^\circ)$ has conditional mean

[6] T2: along a 12-point homotopy the realized gap $g(a) - g(K_F^*) \geq 0$ and $dK_R^*/dC_M \leq 0$ at every node; and $g(\cdot; K_R)$ is non-increasing on the full support for every sampled K_R.

at least the midpoint, $\mu^\circ_{\text{mid}} \geq K^\circ_F/2$, and since $\varphi^\circ - q^\circ > 0$, $\mathbb{E}[\text{fossil}^\circ] \geq K^\circ_F(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K^\circ_F/2$. Subtracting gives $\Delta \geq \Delta_{\text{LB}}$. The capacity channel $K^*_R > K^\circ_R$ (Lemma 4) is what makes the renewable term $\frac{C_M}{R-c} \cdot \frac{a+K^*_R}{2}$ large: a strictly larger K^*_R widens the FCL surplus zone and lowers $\mathbb{E}[\text{fossil}^*]$. Hence the listed hypotheses—endpoint shift-ratio (channel) plus the two-zone bound—are jointly sufficient for $\Delta > 0$.[7] □

Whether the bound is *activated* is the real question. We resolve it with two proved facts—a fossil floor and an exact impossibility—plus a numerical benchmark. We stay in the large-FCL regime throughout (Role 1's mandate); the small-FCL regime is Role 4's and we make no claim about it.

Lemma 5 (Fossil floor)**.** *In the large-FCL regime ($K^*_M \geq K^*_R$), using only the cutoff φ_1,*

$$\mathbb{E}[\textit{fossil}^*] \;\geq\; K^*_F\,(1-\varphi_1) = K^*_F\,\frac{R-c-C_M}{R-c} \qquad \textit{and} \qquad \mathbb{E}[\textit{fossil}^*] \geq K^*_M - K^*_R. \tag{14}$$

The first holds because $F \leq F(a) = \varphi_1$ on $(-K^*_M, a)$, so $\mathbb{E}[\textit{fossil}^*] = \int_{-K^*_M}^{a}(1-F) \geq (a+K^*_M)(1-\varphi_1) = K^*_F(1-\varphi_1)$; the second because $F \equiv 0$ on $(-K^*_M, -K^*_R)$. Physically the second is the floor*: the FCL exceeds the maximal renewable surplus $\leq K^*_R$, so an $K^*_M > K^*_R$ FCL draws at least $K^*_M - K^*_R$ fossil even at deepest surplus.*[8]

The floor lower-bounds $\mathbb{E}[\text{fossil}^*]$ but does not by itself sign $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$, because the channel $K^*_R > K^\circ_R$ pushes the other way. The next result explains why *no* distribution-free argument can sign it—the sign is irreducibly distributional.

Proposition 2 (The carbon sign is irreducibly distributional)**.** *Hold the equilibrium capacities $(K^*_F, K^*_M, K^*_R, K^\circ_F)$ and the surplus probability q° fixed. Reading off (11)—$\mathbb{E}[\textit{fossil}^*]$ is linear in μ^*_{low} and $\mathbb{E}[\textit{fossil}^\circ]$ in μ°_{mid}—Δ is* affine *in the two conditional means $\mu^*_{\text{low}} \in (-K^*_R, a)$ and $\mu^\circ_{\text{mid}} \in (0, K^\circ_F)$, with coefficients fixed by those capacities and cutoffs; the means themselves are* not *determined by capacities and cutoffs. Writing Δ_0 for the value at the interval midpoints (the uniform shape),*

$$\Delta_0 := K^\circ_F\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] - \Big(K^*_F - \tfrac{C_M}{R-c}\cdot\tfrac{a+K^*_R}{2}\Big), \tag{15}$$

the two-zone tilts sandwich Δ around it—$\Delta \geq \Delta_0$ under Cond. 3(a,b), $\Delta \leq \Delta_0$ under the reverse tilt (a′,b′), equality at the uniform. (Note Δ_0 is exactly the keystone bound Δ_{LB} of (13): the only "sufficient-for-fall" statement is $\Delta_0 > 0$, and Cond. 3 is precisely the tilt giving $\Delta \geq \Delta_0$.) By the tight bounds of Theorem 5 (proved below from (11)), as the means sweep their intervals Δ ranges over $[\Delta_{\min}, \Delta_{\max}]$, and at the equilibrium capacities this range straddles zero*: $\Delta_{\min} < 0 < \Delta_{\max}$ (e.g. at the representative optimum, with q° fixed, $[\Delta_{\min}, \Delta_{\max}] = [-6.43, +3.06]$). Consequently* ***the equilibrium capacities and cutoffs together do not determine the sign of*** *Δ—a function of those quantities alone cannot equal* $\operatorname{sign}\Delta$*, since both signs occur at the same capacities and cutoffs for different admissible conditional laws. Only the conditional means break the tie, which is why Week-1's conjecture admits no proof from the distribution-free structure alone.*[9]

Proof. The affine representation is immediate from (11); the midpoint bounds for monotone densities are the standard least-favorable fact (a density non-increasing on $[\ell, u]$ has mean $\leq (\ell+u)/2$), and

[7] T3: the sufficient bound Δ_{LB} is computed at every interior optimum scanned; the algebra $\mu^*_{\text{low}} \leq (a - K^*_R)/2$, $\mu^\circ_{\text{mid}} \geq K^\circ_F/2$ under (a),(b) is checked directly.

[8] The fossil formula already credits surplus absorption—a state-by-state simulation serving FCL from surplus first matches $\int_{-K^*_M}^{a}(1-F)$ to ≈ 6 digits (T3, 6.18324 vs 6.18325); the floor is the *residual* after that credit.

[9] Caveat (as in Theorem 5): the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F. The claim is under-determination of sign Δ by capacities and cutoffs over the admissible-conditional-law class, not that an equilibrium attains the extremes.

reversing each monotonicity reverses the corresponding inequality, giving the sandwich. Since the two extremal tilts are both admissible at the *same* cutoffs, the cutoffs cannot decide sign Δ.[10] □

So the carbon question reduces to the sign of the cutoff-computable benchmark Δ_0 plus a distributional tilt. We settle both numerically.

Proposition 3 (Numerical resolution, large-FCL)**.** *Proposition 2 makes the carbon sign a genuinely distributional quantity—so its resolution* must *be numerical, and we make that resolution exhaustive. For the non-power-law Beta scale-mixture family ($\tilde{D} \sim \bar{D}\,\mathrm{Beta}(\alpha, \beta_D)$, $\tilde{S} \sim \mathrm{Beta}(\alpha_S, \beta_S)$), a sweep of more than* 250 *parameter cells (demand skews α, β_D, capacity factors $\mathbb{E}[\tilde{S}]$ up to* 0.8*, the $C_R \times C_M$ grid) yields, at* every *large-FCL interior optimum located:* (i) *the floor* (14) *holds;* (ii) *the uniform-shape benchmark is already negative, $\Delta_0 < 0$ ($\Delta_0^{\max} \approx -1.09$); and* (iii) *the realized $\Delta < 0$ ($\Delta^{\max} \approx -1.4$)—the distributional tilt* reinforces *the negative benchmark rather than reversing it. Given the impossibility result, this is the strongest statement the structure admits short of a distribution-specific computation: we therefore state it as the* (numerically established) *resolution, with the residual "for all F" universal a conjecture by necessity, not by lack of proof effort. Carbon strictly falls only at the cheap-renewable knife-edge* corner $K_R^* \to \infty$ *(power-law, the window $(C_R^\circ, C_R^\dagger)$ of Week-1 Theorem 4(ii)), where $\mathbb{E}[\textit{fossil}^*] = 0 < \mathbb{E}[\textit{fossil}^\circ]$.*

Remark 3 (Status, and an earlier slip corrected). Week 1 conjectured carbon falls at an interior optimum but never exhibited one. The treatment here is, we believe, as far as the question can be pushed analytically: Lemma 5 proves the structural floor; Proposition 2 proves that the sign is *not* cutoff-decidable (so the conjecture admits no distribution-free proof—a genuine obstruction, not a gap in effort); and Proposition 3 resolves the residual distributional question numerically, finding the benchmark Δ_0 already negative and the tilt reinforcing it at every scanned optimum. Interior carbon reduction is, at best, rare and unwitnessed; the only clean strict fall is the corner. (An earlier draft mis-stated the mechanism as "FCL draws fossil on surplus"; the fossil formula already credits surplus absorption—the correct obstruction is the floor.) The small-FCL regime ($K_M^* < K_R^*$) is outside Role 1's large-FCL mandate and is left to Role 4.

5. Connectedness of the carbon-reduction set

Let $\mathcal{S} := \{C_R \geq 0 : \Delta(C_R) \geq 0\}$ be the set of renewable costs at which the program reduces carbon. Week 1 (Thm. 4) proved $\mathcal{S}$ bounded above for every F; it is non-empty via the cheap-renewable *corner* of the power-law family (Proposition 3: on $(C_R^\circ, C_R^\dagger)$, $K_R^* \to \infty$ gives $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ]$). On the Beta-mixture *interior*, by contrast, $\mathcal{S}$ may be empty (the same proposition). We now give the condition under which $\mathcal{S}$ is a single interval.

Condition 4 (Monotone renewable shadow value, MRSV)**.** At every C_R where the program is carbon-neutral, $\Delta(C_R) = 0$, the saving strictly down-crosses: $\Delta'(C_R) < 0$. Equivalently, at such a C_R the FCL fossil output rises strictly faster in C_R than the no-FCL output, $\frac{d}{dC_R}\mathbb{E}[\text{fossil}^*] > \frac{d}{dC_R}\mathbb{E}[\text{fossil}^\circ]$.

This is a bare sufficient condition; we do *not* have a primitive justification for it. One might guess it follows from a shadow-value ordering $\sigma^* > \sigma^\circ$ ($\sigma := -d\,\mathbb{E}[\text{fossil}]/dK_R$), but that guess is *false*: numerically $\sigma^* < \sigma^\circ$ (the no-FCL renewables carry the larger marginal fossil impact, since they displace fossil directly while the FCL program's renewables partly feed computing). So Condition 4 is genuinely a hypothesis, not a derived fact—and, as the next theorem shows, it is also *not needed* for the family where carbon actually falls.

Theorem 3 (Single-interval carbon-reduction set)**.** (i) *$\mathcal{S}$ is bounded above: there is $B_0 < \infty$ with $C_R > B_0 \Rightarrow K_R^* = K_R^\circ = 0$ and $\Delta(C_R) = -U_0 < 0$, where $U_0 := \left(\mathbb{E}[\textit{fossil}^*] - \mathbb{E}[\textit{fossil}^\circ]\right)\big|_{K_R^* = K_R^\circ = 0} >$*

[10] T3 checks $\Delta \geq \Delta_0$ holds at the scanned optima with the realized (reverse-tilted) densities giving $\Delta \leq \Delta_0$, and that Δ_0 is computed from cutoff+capacity data.

0 *is the pure utilization penalty—with no renewables to offset it, the* FCL *only adds fossil.* (ii) (Power-law/scale family—exact, unconditional.) $\mathcal{S} = [0, C_R^\dagger]$*, a single bounded interval, with the strict-reduction core* $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ *the open window of Week-1 Theorem 4(ii) (closed-form thresholds* $C_R^\circ < C_R^\dagger$*). In particular* $\mathcal{S} \neq \varnothing$*.* (iii) (General F, conditional extension.) Δ *is* C^1 *in* C_R *on the large-*FCL *interior (Corollary 1); under Condition 4,* Δ *is then single-crossing on* $\{C_R : \Delta \geq 0\}$*, so any interior reduction set is a single interval. (By Proposition 3 there is no interior reduction in the scanned family, so (iii) is the empty-or-corner case; it matters only if some* F *outside the scan reduces carbon at an interior optimum.)*

Proof. (i) is Week-1 Theorem 4(i): above the finite renewable-benefit ceiling B_0 the renewable FOC has the corner $K_R^* = K_R^\circ = 0$, leaving only the distribution-free utilization effect, so $\Delta = -U_0 < 0$. (ii) is the Remark below (Week-1 closed forms). For (iii), Δ is C^1 on the large-FCL interior by Corollary 1; we show $\{C_R : \Delta \geq 0\}$ restricted to that region is convex. Suppose not: then there are $C_R^{(1)} < C_R^{(2)} < C_R^{(3)}$ with $\Delta(C_R^{(1)}) \geq 0$, $\Delta(C_R^{(2)}) < 0$, $\Delta(C_R^{(3)}) \geq 0$. By the intermediate value theorem Δ has a zero in $(C_R^{(2)}, C_R^{(3)})$; let z be the smallest such zero. On $[C_R^{(2)}, z)$, $\Delta < 0$ (else there would be an earlier zero), and $\Delta(z) = 0$; hence Δ is non-decreasing as it reaches z from the left, $\Delta'(z) \geq 0$. This contradicts Condition 4, which forces the strict down-crossing $\Delta'(z) < 0$ at every zero z. So no interior dip exists and $\mathcal{S}$ is convex, i.e. an interval; boundedness (i) makes its right endpoint $C_R^\dagger = \sup \mathcal{S}$ finite, and strict down-crossing rules out any second crossing.[11] □

Remark 4 (The power-law set is exactly a single interval, unconditionally). For the family where the actual carbon-reduction set is non-trivial—the power-law/scale family—no condition is needed: by Week-1's explicit-$C_R^\dagger$ proposition, on $(C_R^\circ, C_R^\dagger)$ one has $K_R^* = \infty$, $\mathbb{E}[\text{fossil}^*] = 0$, and K_R° finite with $\mathbb{E}[\text{fossil}^\circ] > 0$, so $\Delta > 0$ *exactly* there, while $\Delta < 0$ for $C_R > C_R^\dagger$ and $\Delta = 0$ for $C_R \leq C_R^\circ$ (both grids hit the unbounded-renewable corner, so both fossil outputs vanish and there is no *strict* reduction). Hence $\mathcal{S} = \{\Delta \geq 0\} = [0, C_R^\dagger]$ is a single bounded interval by the closed-form thresholds, with *no* appeal to Condition 4; the *strict*-reduction set $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ is its open core, and the flat $[0, C_R^\circ]$ is the degenerate "both fully decarbonized" piece. (This is the "carbon strictly falls only in the corner" statement of the abstract: strict fall happens exactly on $(C_R^\circ, C_R^\dagger)$, the corner window.) The MRSV machinery is the general-F extension, and—given Proposition 3, which finds no interior reduction in the scanned family—it is the only place a general-F reduction set could be non-trivial off the corner. The C^1-ness that (iii) needs is now *proved* on the large-FCL interior (Corollary 1), so only Condition 4 itself is a hypothesis, and it is close to its own conclusion ("strictly down-crosses at every zero" *is* single-crossing). We make no claim that it holds: as noted above, the natural shadow-value justification fails ($\sigma^* < \sigma^\circ$, verified in §7). This costs the deliverable nothing, because part (ii)—the only family where carbon actually falls—gives the single interval *exactly*, needing neither Condition 4 nor C^1-ness. Part (iii) is included only to record what a general-F extension would require, and is vacuous in the scanned family (no interior reduction, Proposition 3).

6. Sharpness: the necessary-and-sufficient channel condition and tight bounds

6.1. **The sharp channel condition is a path integral.**

Week 1 used pointwise-monotone g as a *sufficient* condition for $K_R^* > K_R^\circ$. The sharp condition is integral, not pointwise.

Theorem 4 (Necessary and sufficient channel condition)**.** *On the open large-*FCL *interior interval where Condition 1 holds (*$\mathcal{H} \prec 0$*), write* $\lambda(C_M) := -dK_R^*/dC_M = \kappa(C_M)\, f(K_F^*) f(a)\, [\, g(a) - g(K_F^*)\,]$ *with* $\kappa(C_M) = (\beta + p - R)(R - c)/|\det \mathcal{H}| > 0$*. For any compact subinterval* $[C_M, C_M']$ *of that open*

[11] T4: sweeping C_R on the witness family, $\Delta(C_R)$ has exactly one sign change ($+ \to -$), is non-increasing where $\Delta \geq 0$, and is negative at the largest C_R.

interval,

$$K_R^*(C_M) - K_R^*(C_M') = \int_{C_M}^{C_M'} \lambda(t)\, dt, \tag{16}$$

and, letting $C_M' \uparrow \bar{C}_M$ *where* $K_R^*(C_M') \to K_R^\circ$ *by continuity (Week 1),* $K_R^*(C_M) > K_R^\circ$ if and only if $\int_{C_M}^{\bar{C}_M} \lambda > 0$. *Pointwise* $g(a) \geq g(K_F^*)$ *(Condition 2) makes* $\lambda \geq 0$ *and is therefore* sufficient*; it is* not necessary*, since the integral can be positive while the integrand is negative on part of the path.*

Proof. On the open interior interval Condition 1 gives $\mathcal{H} \prec 0$, so by the implicit function theorem $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ is C^1 there, and $\lambda = -dK_R^*/dC_M$ is *continuous*. On a compact subinterval $[C_M, C_M']$ a continuous derivative is integrable, so K_R^* is C^1 (a fortiori absolutely continuous) and the fundamental theorem of calculus gives (16) exactly. The factorization of λ is Lemma 3 divided by $|\det \mathcal{H}| > 0$. As $C_M' \uparrow \bar{C}_M$ the first FCL unit's marginal profit vanishes, $K_M^* \to 0$, and the system collapses to the no-FCL optimum, $K_R^*(C_M') \to K_R^\circ$ (Week 1, continuity at the adoption boundary). At that boundary $a = K_F^* - K_M^* \to K_F^*$, so the cofactor bracket $g(a) - g(K_F^*) \to 0$ while $|\det \mathcal{H}| = |\det \mathcal{M} \cdot \Psi'|$ stays bounded away from 0 ($\det \mathcal{M} \to (R-c)(\beta+p-R)f(K_F^*)^2 > 0$ and $\Psi' < 0$ on the interior); hence the cofactor-over-determinant ratio λ extends *continuously* to $\bar{C}_M$ with $\lambda(\bar{C}_M^-) = 0$, the integral $\int_{C_M}^{\bar{C}_M} \lambda$ is *proper*, and by (16) it equals $\lim_{C_M' \uparrow \bar{C}_M} [K_R^*(C_M) - K_R^*(C_M')] = K_R^*(C_M) - K_R^\circ$. Positivity of $K_R^*(C_M) - K_R^\circ$ is then equivalent to positivity of $\int_{C_M}^{\bar{C}_M} \lambda$. Sufficiency of Condition 2 is $\lambda \geq 0$ pointwise; non-necessity is the elementary fact that an integral of a sign-changing integrand can be positive.[12] □

6.2. **The two-sided bounds are tight.**

Theorem 5 (Tight magnitude bounds)**.** *Fix the equilibrium capacities and* q°*, and range over all conditional laws consistent with the distribution-free cutoffs (these laws need not all be realized by an equilibrium* F*; the extremizers below are degenerate, not equilibrium laws). With* Δ *written as the affine map* (11) *of the conditional means* $\mu_{\text{low}}^* \in (-K_R^*, a)$, $\mu_{\text{mid}}^\circ \in (0, K_F^\circ)$,

$$\Delta_{\min} := K_F^\circ(1-\varphi^\circ) - K_F^* \;\leq\; \Delta \;\leq\; K_F^\circ(1-q^\circ) - K_F^* + \tfrac{C_M}{R-c}(a + K_R^*) := \Delta_{\max}, \tag{17}$$

and both bounds are the exact sup/inf of the affine map at the fixed q°*, attained in the limit:* $\Delta_{\min}$ *as the* FCL *conditional law concentrates at the upper endpoint* ($\mu_{\text{low}}^* \uparrow a$) *and the no-*FCL *law at the lower endpoint* ($\mu_{\text{mid}}^\circ \downarrow 0$)*;* $\Delta_{\max}$ *as the laws concentrate oppositely* ($\mu_{\text{low}}^* \downarrow -K_R^*$, $\mu_{\text{mid}}^\circ \uparrow K_F^\circ$)*. Here* $\mathbb{E}[\textit{fossil}^\circ]$ *rises to its maximum* $K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ = K_F^\circ(1-q^\circ)$ *as* $\mu_{\text{mid}}^\circ \uparrow K_F^\circ$*, with* q° *held at its value.*

Proof. Δ is affine and monotone in each conditional mean: $\partial\Delta/\partial\mu_{\text{low}}^* = -C_M/(R-c) < 0$ and $\partial\Delta/\partial\mu_{\text{mid}}^\circ = \varphi^\circ - q^\circ > 0$. A conditional mean ranges over the open interval between its endpoints, with the endpoints approached by laws concentrating there (e.g. a two-point law placing mass $\to 1$ at the endpoint while preserving the cutoff probability). Plugging the extreme means into (11) gives (17), and the attaining sequences are the concentrating laws just named; the bounds are therefore the exact sup/inf, i.e. tight.[13] □

7. Numerical validation

All identities are checked in `verify_week2.py` against adaptive quadrature of the profit integral (1), on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$. T1 confirms the recursive triangle (interiority-only (K_F, K_M)-block, scalar Schur, $\mathcal{H} \prec 0$, unique Ψ zero); T2 the uniform endpoint

[12] T5: on the interior band $\int_{C_M}^{C_M'} (-dK_R^*/dC_M)\, dt$ matches $K_R^*(C_M) - K_R^*(C_M')$ to $\sim 2\%$ (4.21 vs 4.13); the sufficient-not-necessary logic is shown by a sign-mixed integrand with positive integral. A *realized* F with $\lambda < 0$ on a sub-path is Role 3's reversed-channel counterexample, outside this deliverable.

[13] T5b: $\Delta(\mu_{\text{low}}^* = a, \mu_{\text{mid}}^\circ = 0) = \Delta_{\min}$ and $\Delta(-K_R^*, K_F^\circ) = \Delta_{\max}$ exactly, and the realized Δ lies strictly inside.

inequality along the homotopy; T3 the exact endogenous identity, the sufficient-bound algebra, and the Proposition 3 scan ($\Delta < 0$ at every large-FCL interior optimum, $\Delta^{\max} \approx -1.4$); T4 the corner non-emptiness $C_R^{\circ} < C_R^{\dagger}$, the large-cost anchor, and the single interval; T5 the FTC/path-integral identity and the tight bounds $\Delta_{\min} \leq \Delta \leq \Delta_{\max}$ attained at the endpoints. The suite reports a single pass count.